

Ahmed Kheder

📍 Cairo, Egypt ✉ ahmed.kheder2002@gmail.com ☎ +20 109 848 1767 in ahmed-kheder1
🔗 AhmedKhadr1

Profile

AI & Robotics enthusiast with a Computer Engineering background, skilled in Python and TensorFlow. Leverage data-driven methods and predictive models to develop intelligent systems that address real-world problems. Precise, curious, and committed to continual innovation

Education

University of Ain Shams Cairo, Egypt
BSc in Computer and Systems Engineering Sep 2021 – Present

- **GPA:** 3.52/4.00
- **Key Modules:** Machine Learning, Deep Learning, Operating Systems, Database Systems, Data Structures & Algorithms, Advanced Programming, Artificial Intelligence, Embedded Systems, and Control Engineering

Projects

Motor Imagery Classification | *Python, Pandas, Numpy* [GitHub Repository](#)

- Segmented continuous Muse-S motor-imagery EEG into epoch-labelled DataFrames enriched with subject / session metadata
- Built an automated MNE-Python pipeline band-pass + notch filters, ICA artefact removal, CAR, z-score normalisation to clean raw signals
- Applied offline augmentation (Gaussian noise, temporal shifts, window jitter) and packaged data in efficient PyTorch Dataset/DataLoader
- Implemented and fine-tuned EEGNet-8-2 CNN and Bi-LSTM architectures in PyTorch/TensorFlow with reproducible training and checkpointing

San Francisco Crime Classification | *Python, Pandas, Numpy* [GitHub Repository](#)

- Cleaned and engineered crime data: parsed temporal features, removed spatial outliers/duplicates, one-hot-encoded categorical variables, and standard-scaled numerics into a model-ready matrix
- Trained tuned Random Forest, LightGBM, and Logistic Regression classifiers, benchmarking performance with multi-class ROC curves
- Produced spatial crime visualisations

Professional Development

Machine Learning Specialization | *DeepLearning.AI & Stanford University*

- Supervised ML: Regression & Classification
- Unsupervised Learning, Recommenders, Reinforcement Learning

AI Data Science and Machine Learning Diploma | *WE Telecom Egypt*

- Machine learning: data cleaning, feature engineering, model tuning
- Deep-learning foundations with TensorFlow (CNNs, RNNs)
- Capstone: built image-classification model

Python for Everybody | *University of Michigan*

- Python Basics | Data Collection & Processing with Python
- Python Functions, Files & Dictionaries | Python Classes & Inheritance

Technical Skills

Languages: C/C++, Python, Java, SQL, Assembly (ARM)

Machine Learning & AI: XGBoost, Scikit-learn, Tensorflow, PyTorch, RegEx

Data Manipulation and Visualization: NumPy, Pandas, Matplotlib, Seaborn

Tools & Technologies: MongoDB, MySQL, Qt, Git/GitHub, MATLAB